Dataset Analysis and Insights

March 19, 2025

sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address your request, I will first load the dataset and examine its structure, key characteristics, and potential insights. Once I understand the dataset, I will be ready to address any specific analytical question you have. Below is the step-by-step process:

Step 1: Load and Explore the Dataset

I will load the dataset using Python's `pandas` library and inspect its structure, including column names, data types, and a few sample rows.

```python
import pandas as pd

# Load the dataset
file_path = 'sicor_municipal_contracts.csv'
data = pd.read_csv(file_path)

# Display basic information about the dataset
print("Dataset Overview:")
print(data.info())

# Display the first few rows of the dataset
print("\nFirst 5 Rows of the Dataset:")
print(data.head())
```

Step 2: Identify Key Characteristics

After loading the dataset, I will summarize its key characteristics:

- **Columns** : What variables are included in the dataset?
- **Data Types** : Are the columns numeric, categorical, or textual?
- **Missing Values** : Are there any missing or null values in the dataset?
- **Summary Statistics** : For numeric columns, what are the mean, median, min, max, etc.?

```python
# Check for missing values
print("\nMissing Values per Column:")
print(data.isnull().sum())

# Summary statistics for numeric columns
print("\nSummary Statistics for Numeric Columns:")
print(data.describe(include='all'))
```

Step 3: Address Specific Analytical Questions

Once you provide a specific analytical question, I will:

1. Extract and process the relevant data from the dataset.
2. Perform the necessary calculations or analyses.
3. Provide a clear, concise answer to your question along with the Python code used.

Next Steps

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Please provide a specific analytical question related to the dataset (e.g., "What is the total value of contracts awarded in each municipality?" or "Which contractor received the highest number of contracts?"). I will then proceed with the analysis and deliver the requested insights.

What is the proportion of investment value per month?

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To calculate the proportion of investment value per month, we need to:

1. Group the data by month.
2. Sum the investment values for each month.
3. Calculate the proportion of each month's total investment value relative to the overall total investment value.

From the dataset, the relevant columns appear to be:

- Column 0: Month (assumed to represent the month of the contract).
- Column 10 or similar: Investment value (likely representing the contract value or monetary amount).

Here is the Python code to compute the proportions:

```python
import pandas as pd

# Load the dataset
file_path = 'sicor_municipal_contracts.csv'
data = pd.read_csv(file_path, header=None)

# Rename columns based on assumptions
data.rename(columns={
    0: 'month',
    10: 'investment_value'
}, inplace=True)

# Group by month and sum the investment values
monthly_investment = data.groupby('month')['investment_value'].sum()

# Calculate the total investment value
total_investment = monthly_investment.sum()

# Calculate the proportion of investment value per month
monthly_proportions = monthly_investment / total_investment

# Display the results
print("Monthly Investment Proportions:")
print(monthly_proportions)
```

Explanation of the Code:

1. **Column Assumptions** :
   - The first column (`0`) is assumed to represent the month.
   - The eleventh column (`10`) is assumed to represent the investment value.
2. **Grouping and Summing** :
   - We group the data by the `month` column and sum the `investment_value` for each month.
3. **Proportion Calculation** :
   - The proportion for each month is calculated by dividing the monthly total by the overall total investment value.
4. **Output** :
   - The result is a series where the index represents the month, and the values represent the proportion of investment for that month.

Example Output:

If the dataset contains the following aggregated monthly investment values:

- Month 1: $500,000
- Month 2: $300,000

- Month 3: $200,000

And the total investment is $1,000,000, the proportions would be:

- Month 1: 0.5 (50%)
- Month 2: 0.3 (30%)
- Month 3: 0.2 (20%)

Note:

- Ensure that the column indices (`0` for month and `10` for investment value) are correct based on the actual dataset structure. If these assumptions are incorrect, please provide clarification.
- If there are additional complexities (e.g., missing or zero values in the investment column), further preprocessing may be required.